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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-58

Charging Stations

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Recap

- AC type2 charging systems
- CCS2 DC charging systems
- DC001 DC charging systems

Recap: Charging Levels

- In India, the charging levels are defined as

S. No	Charging Level	Voltage (V)	Power (kW)
1	Level 1 (AC)	230	<=3.5 kW
2	Level 1 (DC)	>=48	<=15 kW
3	Level 2 (AC)	380-400	<=22 kW
4	Level 3 (AC)	200-1000	> 22 kW
5	Level 3 (DC)	200-1000	up to 400 kW

230 × 15 = 3.45 kW

72 × 200 = 14400 W

Source: https://e-amrit.niti.gov.in/standards-and-specifications

Charging Station

Public charging station (Malls, restaurants, movie halls, public Parking)

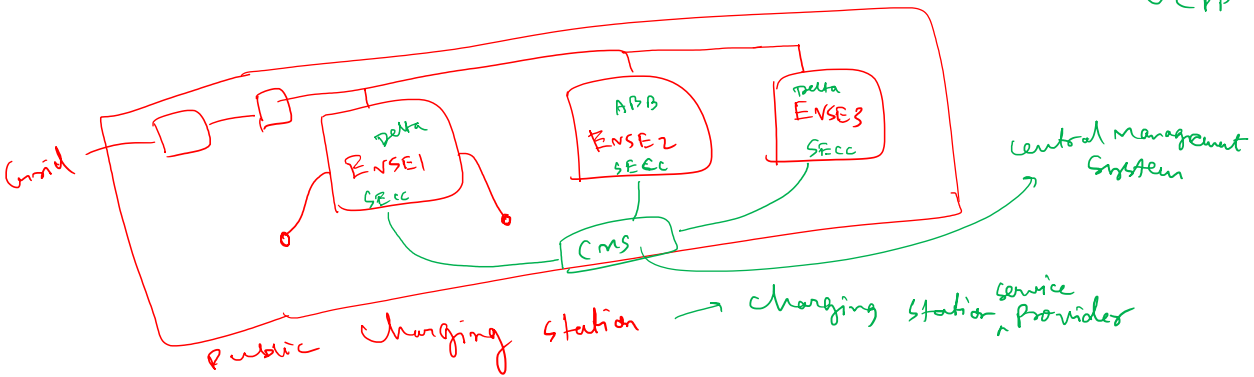
Home charging (EVSE with single connector gun)

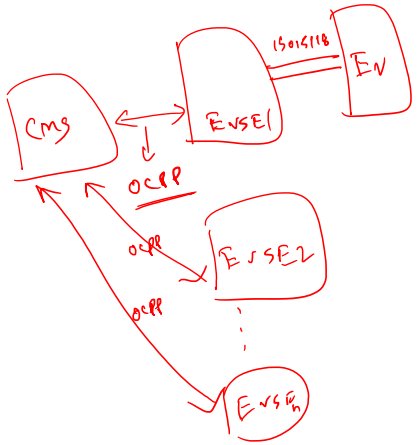
Captive charging station (it is not for public, and 100% used by the institution in which it is installed)

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have its own Transformer, Switchgear, safety features, cabling

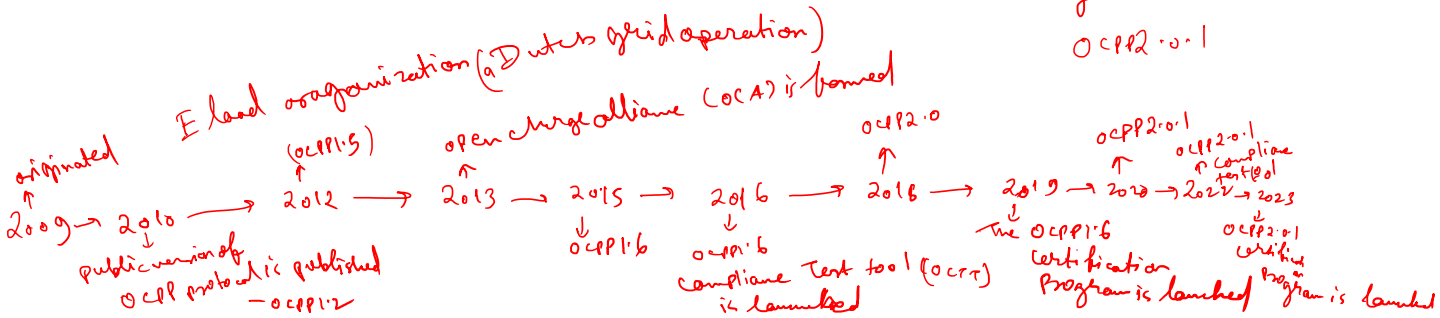
OCPP (Open charge Point Protocol)




CMS
EVSE1
ISO15118
EV
OCPP
OCPP
EVSE2
OCPP
EVSEn

OCPP (Open Charge Point Protocol)

- OCPP provides a standardized protocol for communication between EVSE and central management systems. (CMS)
- The protocol enables the exchange of messages and data between EVSE and central management systems.
- OCPP standard has some key components: OCPP profiles, data types, messages, configuration keys, use case and test case.
- OCPP profile: define how the two entities communicate, including the messages exchanged, the message format, and the specific functionality of each system. (entity)



OCPP

- Message: data unit transmitted between EVSE and CMS during a charging session. It has a specific format and purpose.
- Data type: OCPP profiles utilize particular data types to specify the format and organization of data including Boolean, string, date-time, enumeration, etc.
- Configuration keys: Configuration key specifies different settings and restrictions that manage how the EVSE operates throughout a charging session. This covers charging rates, maximum power limits, and authorization requirements. Every profile contains a collection of predefined configuration keys.
- Use case: A use case describes a particular situation that the protocol is intended to handle. Examples of use cases are start transaction, stop transaction, etc.
- Test cases: Developers rely on them to confirm the proper implementation of the OCPP protocol. Each OCPP profile contains a collection of predefined test cases that help developers validate the behavior of their implementation.

Source: https://www.ampeco.com/guides/complete-ocpp-guide/

Thank You